The Sharp Twelve-Point Bound in Mordell's Distance-Product Problem

Fengcan Jiang's Mordell research project

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Abstract

We give a computer-assisted proof of the sharp twelve-point bound for the product of squared pairwise distances in the complex plane at fixed quadratic mass. The unique maximizing orbit consists of two regular triangles at distinct radii and two further regular triangles sharing an outer radius. A preliminary two-point certificate forces all global maximizers into two radial bands. On those bands a second kernel is made to interpolate the candidate exactly. Fourteen contact neighborhoods are certified by interval Hessian bounds, and a finite Bernstein subdivision certifies the rest with zero slack. The equality conditions of the positive-semidefinite part recover the full geometry.

1 Problem and result

For $z = (z_1, \dots, z_n) \in \mathbb{C}^n$, put

$$Q(z) = \sum_{i=1}^{n} |z_i|^2, \qquad \Delta_n(z) = \prod_{1 \le i < j \le n} |z_i - z_j|^2, \qquad M_n = \max_{Q(z)=n} \Delta_n(z).$$

Ji, Xu, and Yu [1] give sharp results through nine points, including a two-point approach for eight and nine. The twelve-point candidate has three distinct positive radii and requires a kernel whose equality conditions distinguish radial populations as well as relative phases.

Theorem 1 (Twelve-point maximum)**.** *Let z^* be the exact mass-eleven candidate specified in Section 4. Then*

$$M_{12} = \left(\frac{12}{11}\right)^{66} \Delta_{12}(z^*) \simeq 2.599464093232166 \times 10^{14}.$$

Every maximizer is a rotation and relabeling of $\sqrt{12/11}\, z^*$. The three radii have populations $3, 3, 6$. The six-point layer is the union of two regular triangles at the same radius.

The proof has three finite certificate stages. A rational Ward kernel supplies a global deficit budget, which becomes a radius restriction through affine row inequalities. A candidate-saturating kernel is reconstructed from the exact root and an invertible contact minor. Finally, local and complementary boxes prove the sharp pointwise inequality. We give the mathematical implications and the acceptance tests below. The fixed inputs and full replay programs are available in the fixed certificate archive [2].

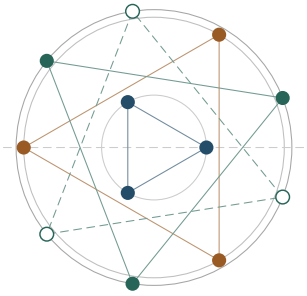


Figure 1: The candidate in the mass-eleven normalization. The solid and dashed green triangles share the outer circle. The drawing uses rounded coordinates.

2 Stationarity and the kernel summation identity

2.1 Normalization and full stationarity

Write $N = \binom{n}{2}$. Multiplying the quadratic mass by $\eta > 0$ multiplies the optimal product by η^N. We use the mass $n-1$ normalization throughout the proof, so the multiplier is $\lambda = n/2$. A global maximum exists by compactness and has positive product, since a regular polygon is feasible after scaling. Its points are distinct. Translating to the centroid preserves the product and reduces the mass unless the centroid is zero. Rescaling to the prescribed mass would then strictly increase the product. Thus every global maximizer is centered.

The real Lagrange equations for $\log \Delta_n$, in complex notation, are

$$\mathcal{F}_i(z) = \sum_{j \neq i} \frac{1}{z_i - z_j} - \lambda \overline{z_i} = 0, \qquad \lambda = \frac{N}{Q(z)}. \tag{1}$$

Indeed, multiplying by z_i and summing counts each unordered pair once. We also use that a constrained global maximizer at mass $n-1$ maximizes $\Delta_n(z)e^{-\lambda Q(z)}$ on the full configuration space. At a given positive mass q the product is bounded by its constrained maximum at mass $n-1$ multiplied by $(q/(n-1))^N$. The scalar function $q^N e^{-\lambda q}$ is maximized at $q = n-1$.

2.2 Positive Gram blocks and zero-sum Ward corrections

Put $u = |z|^2$, $v = |w|^2$, $t = \mathrm{Re}(z\bar{w})$ and $d = u + v - 2t$. Set

$$H_k(u,v,t) = \mathrm{Re}((z\bar{w})^k), \qquad H_0 = 1, \quad H_1 = t, \quad H_k = 2tH_{k-1} - uvH_{k-2} \quad (k \geq 2).$$

For real polynomial feature vectors f_k and real symmetric blocks $Q_k \succeq 0$ the base kernel

$$K(u,v,t) = \sum_{k=0}^{11} H_k(u,v,t) f_k(u)^T Q_k f_k(v) \tag{2}$$

is positive semidefinite. For any real weights on any finite point set its quadratic form is a sum of squared moment norms. In particular, with $U_k = \sum_i z_i^k f_k(|z_i|^2)$ and unit weights,

$$\sum_{i,j} K_{ij} = \sum_{k=0}^{11} U_k^* Q_k U_k \geq 0. \tag{3}$$

For the corrections use the raw feature vector $\phi(u) = (1, u-1, \dots, (u-1)^4)^T$ and raw moments $M_k = \sum_i z_i^k \phi(|z_i|^2)$. These differ from the feature moments U_k. Let $p_k = \sum_i z_i^k$, with $p_0 = n$. Multiplication of (1) by z_i^{k+1} and summation give

$$\lambda \sum_i |z_i|^2 z_i^k = \left(n - \frac{k+1}{2}\right) p_k + \frac{1}{2} \sum_{r=1}^{k-1} p_r p_{k-r}. \tag{4}$$

For each unordered pair this follows by expanding the difference quotient $(z_i^{k+1} - z_j^{k+1})/(z_i - z_j)$. Equivalently the right-hand side is $\frac{1}{2}\sum_{r=0}^{k}(p_r p_{k-r} - p_k)$, obtained by subtracting the diagonal from the ordered sum. For $n = 12$ and $Q = 11$, $p_1 = 0$ and the first two coordinates of M_1 vanish. Moreover $v_k^T M_k = 0$ for $k = 0, 2, 3$, with

$$v_0 = (1/12, 1, 0, 0, 0)^T, \qquad v_2 = (-3/4, 1, 0, 0, 0)^T, \qquad v_3 = (-2/3, 1, 0, 0, 0)^T. \tag{5}$$

Here $k = 0$ uses the exact mass alone. For $k = 2, 3$, the coefficients follow from $\sum u_i z_i^2 = 7p_2/4$ and $\sum u_i z_i^3 = 5p_3/3$.

Choose A_1 real symmetric with zero lower-right 3×3 block and choose $A_k = v_k a_k^T + a_k v_k^T$ for $k = 0, 2, 3$. The first choice has nine parameters, and each of the others has five. Define

$$J(u, v, t) = \sum_{k=0}^{3} H_k(u, v, t) \phi(u)^T A_k \phi(v). \tag{6}$$

Then $\sum_{i,j} J_{ij} = \sum_{k=0}^{3} M_k^* A_k M_k = 0$ at every full stationary configuration of mass eleven. For the rank-two forms this follows from $M^*(va^T + av^T)M = 2\,\mathrm{Re}(\overline{v^T M}\, a^T M)$. For A_1 it follows from the two zero coordinates. Thus the ordered sum of $K + J$ is nonnegative on full stationary configurations, including every global maximizer.

2.3 The diagonal correction and the pair deficit

For any real B set $b = (B - 66)/12$. Let $K' = K + J$ and define

$$P_B(u) = \frac{6u + b - \frac{1}{2}K'(u, u, u)}{11}, \qquad h(u, v, t) = P_B(u) + P_B(v) - K'(u, v, t). \tag{7}$$

Each $P_B(u_i)$ appears eleven times in the unordered pair sum. Consequently, at mass eleven,

$$\sum_{i<j} h_{ij} = B - \frac{1}{2} \sum_{i,j} K'_{ij} = B - \frac{1}{2} \sum_{i,j} K_{ij} \quad \text{at full stationary configurations.} \tag{8}$$

The first equality uses the mass alone. The second uses the Ward identities. When $h + \varepsilon \geq \log d$ on a necessary pair domain, the nonnegative pair deficits and the Gram penalties satisfy the exact budget identity

$$\sum_{i<j} (h_{ij} + \varepsilon - \log d_{ij}) + \frac{1}{2} \sum_{k=0}^{11} U_k^* Q_k U_k = B + 66\varepsilon - \log \Delta_{12}(z). \tag{9}$$

We first apply (9) with a small positive ε to localize every maximizer. The final application takes $B = \log \Delta_{12}(z^*)$ and $\varepsilon = 0$.

3 The necessary pair domain and the radial population

Throughout this section x is a global maximizer at mass eleven and $\Delta = \Delta_{12}(x)$. We use its centering and full stationarity from Section 2.

3.1 Strict subset bounds

Use the unconstrained objective $\Delta(x)e^{-6Q(x)}$ from Section 2. Fix a proper subset S of size $m \geq 2$, with centroid a, centered points $y_i = x_i - a$, and mass $q = \sum_{i\in S} |y_i|^2$. Replacing its points by $a + \zeta y_i$ leaves the part of the objective depending on ζ proportional to

$$|\zeta^k P(\zeta)|^2 e^{-6q|\zeta|^2}, \qquad k = \frac{m(m-1)}{2}, \qquad P(\zeta) = \prod_{i\in S,\, j\notin S} (a + \zeta y_i - x_j).$$

The point $\zeta = 1$ maximizes this expression. Thus $f = P/P(1)$ satisfies $f(1) = 1$ and $|f| \leq 1$ on the unit disk. Differentiation in the radial and angular directions at 1 gives $f'(1) = \delta = 6q - k$. Also $P'(0) = 0$, since the coefficient of ζ in the product for each fixed $j \notin S$ is a multiple of $\sum_{i\in S} y_i$.

The polynomial P is nonconstant, since the factors are nonzero polynomials and at least one has nonzero slope. Put $A = f(0)$, so $|A| < 1$. The disk map $g = (f - A)/(1 - \overline{A}f)$ vanishes to order at least two at zero. The Schwarz lemma gives $g(\zeta) = \zeta^2 j(\zeta)$ with $|j| \leq 1$. Because $|j(1)| = 1$, the inward radial limit implies $\mathrm{Re}(j'(1)/j(1)) \geq 0$. Consequently,

$$\delta \geq 2\frac{|1 - A|^2}{1 - |A|^2} \geq 2\frac{1 - |A|}{1 + |A|} > 0. \tag{10}$$

The strict subset bound is $q > m(m-1)/12$. For $0 < \delta < 2$, (10) also gives

$$|P(1)|^2 \leq S(\delta)|P(0)|^2, \qquad S(\delta) = \left(\frac{2 + \delta}{2 - \delta}\right)^2. \tag{11}$$

For two selected points z, w, set $u = |z|^2$, $v = |w|^2$, $t = \mathrm{Re}(z\overline{w})$, and $d = u + v - 2t$. The subset consisting of these two points has mass $d/2 > 1/6$. Deleting the point z leaves an eleven-point subset of mass $11 - 12u/11 > 55/6$. Together with the Gram inequality these give the closed preliminary domain

$$\mathcal{D}_0 = \{(u, v, t) : 0 \leq u, v \leq U_0,\ t^2 \leq uv,\ d \geq 1/3\}, \qquad U_0 = 121/72. \tag{12}$$

The radius and separation inequalities are strict at every maximizer.

3.2 One- and two-point deletion

The finite disk-smearing bound for m points with centered mass q is

$$U_m(q) = m^{m/2}\left(\frac{2q}{m-1}\right)^{m(m-1)/2} e^{-m(m-1)/4}.$$

At mass $m - 1$, write $g(x) = -\log|x|$ and let $D(\nu) = \iint g(x - y)\, d\nu(x)\, d\nu(y)$. The functional $D(\nu) + \frac{1}{2}\int |x|^2\, d\nu$ is minimized among compactly supported probabilities of bounded density by the uniform disk ν_0 of radius $\sqrt{2}$, with value $3/4 - \frac{1}{2}\log 2$. The logarithmic potential $g * \sigma_R$ of a uniform disk of radius R is $-\log R + 1/2 - |x|^2/(2R^2)$ inside and $-\log|x|$ outside. With $R = \sqrt{2}$, the sum $g * \nu_0 + |x|^2/4$ is constant on the disk and at least this constant outside. For $\eta = \nu - \nu_0$,

the quadratic term in the energy comparison is $D(\eta) = (2\pi)^{-1} \int |\nabla(g * \eta)|^2 \geq 0$, by integration by parts using the zero total mass of η. Integrating the disk potential and the confinement term gives the stated minimum.

Replace each point by a uniform disk of radius r, and let ν be the average of these disk measures. The average of g over translations by one disk is at most g, and the same is true for two disks. The self-energy of one disk is $-\log r + 1/4$, and $\int |x|^2 \, d\nu = (m-1)/m + r^2/2$. Subtracting the m self-terms from $m^2 D(\nu)$ thus gives

$$-\log \Delta_m \geq m^2(3/4 - \tfrac{1}{2}\log 2) - \frac{m(m-1)}{2} - \frac{m^2 r^2}{4} + m \log r - \frac{m}{4}.$$

Set $r^2 = 2/m$ and scale from mass $m-1$ to q to obtain $U_m(q)$. This is the finite argument in `asymptotics/proof.tex` of the accompanying repository. It applies to each remaining subset without a stationarity hypothesis. Define rational upper constants by

$$E(x) = \sum_{j=0}^{100} \frac{x^j}{j!}, \qquad C_{11} = \frac{11^5 (33166248/10^7)(1/5)^{55}}{E(55/2)}, \qquad C_{10} = \frac{10^5 (2/9)^{45}}{E(45/2)}.$$

The positive exponential series and $(33166248/10^7)^2 > 11$ give $U_{11}(q) < C_{11} q^{55}$ and $U_{10}(q) < C_{10} q^{45}$ for $q > 0$. Let $V^{(11)}$ be the value of the fixed integer witness in mass eleven. For its original integer points b_i, this is $(132/D)^{66} P_0$, where $D = 12 \sum_i |b_i|^2 - |\sum_i b_i|^2$ and $P_0 = \prod_{i<j} |b_i - b_j|^2$. The b_i are the twelve Gaussian integers in the $n = 12$ entry of the fixed archive's `symmetry-breaking/integer_witnesses.json`.

After one deletion, the interaction constant in (11) is $|P(0)|^2 = (144u/121)^{11}$. For $81/50 \leq u < 121/72$ we have $0 < \delta = 6q - 55 < 2$, and hence

$$\Delta \leq F_{11}(u) = C_{11} q^{55} (144u/121)^{11} S(\delta), \qquad q = 11 - 12u/11.$$

The logarithmic derivative of F_{11} is strictly below $11/u - 144/11 < 0$, using $q' < 0$ and $(2+\delta)^{-1} + (2-\delta)^{-1} \geq 1$. Exact rational evaluation gives $F_{11}(81/50) < (997/1000) V^{(11)}$. Thus every maximizer has

$$u < 81/50. \tag{13}$$

For two deletions let $a = -(z+w)/10$ be the remaining centroid. Now

$$q = 11 - \frac{11(u+v) + 2t}{10}, \qquad \delta = 6q - 45,$$
$$b = |z-a|^2 |w-a|^2 = \frac{(121u + v + 22t)(u + 121v + 22t)}{10^4}.$$

The ten-point subset has $q > 15/2$. When $\delta < 2$, its interaction constant is b^{10}, and the two deleted points contribute d. This gives the necessary rational inequality

$$(2+\delta)^2 C_{10} q^{45} d \, b^{10} \geq (2-\delta)^2 V^{(11)}. \tag{14}$$

For $R = 11 - q$ and $s = d/2$, the centroid decomposition gives $0 \leq s \leq R$ and $|z-a|^2 + |w-a|^2 = (6R - s)/5$. Thus

$$d \, b^{10} \leq 2s \left(\frac{6R - s}{10} \right)^{20} \leq (4R/7)^{21}.$$

The last upper bound occurs at $s = 2R/7$, as the derivative of $s(6R-s)^{20}$ is $(6R-s)^{19}(6R - 21s)$. On $15/2 < q \leq 7573/1000$, the resulting upper bound $F_{10}(q) = C_{10} q^{45} (4(11-q)/7)^{21} S(6q - 45)$ is

increasing, since its logarithmic derivative is at least $45/q - 21/(11-q) + 12 > 0$. Exact arithmetic gives $F_{10}(7573/1000) < (992/1000)V^{(11)}$, so $q > 7573/1000$, or equivalently $11(u+v) + 2t < 3427/100$. Equations (12), (13), and (14), with this last linear bound, give analytic necessary pair restrictions. The one-deletion radius bound will be used for the upper tail of the row cuts. The two-deletion inequalities supply an additional pruning test. The endpoint and witness computations are replayed by `twelve-point-removal-domain/verify_all.py`.

3.3 Ward deficit and the two radial bands

Lemma 2 (Preliminary rational majorant)**.** *The fixed rational kernel K_{W} has twelve positive definite base Gram blocks. Its correction has the Ward form (6). With*

$$B_{\mathrm{W}} = \frac{7726135064073933}{281474976710656}, \qquad c_0 = \frac{1}{25000},$$

its pair function satisfies $h_{\mathrm{W}} + c_0 \geq \log d$ throughout $\mathcal{D}_0$.

The replay reads `twelve-point-ward-upper/kernel.json` and tests every base block by exact rational LDL^T factorization and constructs the raw Ward blocks from their 24 rational parameters. It checks 57 bilinear restrictions on the relevant moment subspaces and reconstructs the polynomial of multidegree at most $(19, 19, 11)$. In the root box $[0, U_0]^2 \times [-U_0, U_0]$ it converts this polynomial to tensor Bernstein coefficients. Every midpoint subdivision rounds its lower coefficients down. Each leaf is accepted by a rational upper tangent to $\log d$, with c_0 added to h_{W}, or by exclusion from the physical domain or the ordered half $u \leq v$. Symmetry covers the other half. This uses the same bounding principle described in Section 6. The archived replay accepts all 268,239 nodes, including 127,832 inequality leaves and 6,288 discarded leaves.

Let $G = h_{\mathrm{W}} + c_0 - \log d$ and let K_{W} denote the base kernel. At a global maximizer, the Ward corrections sum to zero over ordered pairs. The diagonal-corrected summation identity therefore gives

$$\sum_{i<j} G_{ij} + \frac{1}{2} \sum_{i,j} K_{\mathrm{W}}(x_i, x_j) = B_{\mathrm{W}} + 66c_0 - \log \Delta < 66c_0. \tag{15}$$

Here $\Delta \geq V^{(11)}$. A positive-term logarithm expansion after dyadic range reduction gives a rational lower bound for $\log V^{(11)}$ strictly above B_{W}. The pair terms and the ordered sum of each positive Gram block are nonnegative.

For a marked point of radius squared u, its eleven neighbors satisfy $\sum_{j \neq i} v_j = 11 - u$ and $\sum_{j \neq i} t_j = -u$. The certificate partitions $[0, 3/25]$, $[6/25, 47/50]$, and $[71/50, 33/20]$ into 53 closed subintervals with adjacent endpoints. For each subinterval J, fixed rational coefficients a_J, b_J satisfy the continuous inequality

$$G(u, v, t) \geq \frac{1}{4000} + a_J \left(v - \frac{11-u}{11} \right) + b_J \left(t + \frac{u}{11} \right) \quad ((u, v, t) \in \mathcal{D}_0,\ u \in J). \tag{16}$$

The verifier subtracts the two affine terms from the rational polynomial h_{W}. On each accepted box the Bernstein coefficients, rounded down under midpoint subdivision, dominate an upward enclosure of $\log \xi + (d - \xi)/\xi - c_0 + 1/4000$ for a positive rational ξ. Concavity makes this an upper bound for $\log d - c_0 + 1/4000$. Pruned boxes violate the Gram or separation condition. The roles of u and v stay fixed because u marks the row. The 53 trees contain 99,603 nodes, including 46,539 inequality leaves and 3,289 discarded leaves, with every partition closed.

Summing (16) over the neighbors cancels the affine terms, leaving $11/4000 > 66c_0$, in contradiction to (15). The upper tail is excluded by (13) since $81/50 < 33/20$. Every squared radius therefore lies in one of the open bands

$$I = (3/25, 6/25), \qquad O = (47/50, 71/50).$$

3.4 The inner population is three

Let Q_0 be the radial Gram block of K_{W}, and let $m = \sum_i F_0(u_i)$ be its radial moment vector. Its first four coordinates are $\sum_i (u_i^j - \mu_j)$, $1 \le j \le 4$, with the rational centers μ_j read from the fixed kernel. The radial part of (15) gives $m^T Q_0 m < 132c_0$. Delete the first row and column of Q_0 to obtain Q_0^{tail}, solve $Q_0^{\mathrm{tail}} y = (-4, 1, 0, 0)^T$, and put $w = (0, y)^T$. Exact substitution verifies the solution and $w^T Q_0 w < 20$. Set $p(u) = u^3 - 4u^2 + 4\mu_2 - \mu_3$. The difference between $w^T Q_0 m$ and $\sum_i p(u_i)$ is $(11 - 12\mu_1)(Q_0 w)_1$, whose absolute value is strictly below $1/10000$ by rational computation. Thus

$$\left| \sum_i p(u_i) \right| < \frac{1}{10000} + \sqrt{20 \cdot 132c_0} < \frac{41}{100}.$$

The following lower bounds are verified by nonnegative cubic Bernstein coefficients on 64 equal subintervals of each indicated band, for each sign.

	$u \in \overline{I}$	$u \in \overline{O}$
$p(u)$	$347/100 - (26/5)u$	$509/100 - (26/5)u$
$-p(u)$	$-199/50 + (531/100)u$	$-263/50 + (531/100)u$

If the inner population is $k \le 2$, the first row and $\sum_i u_i = 11$ give $\sum_i p(u_i) \ge (388 - 162k)/100 \ge 16/25$. If $k \ge 4$, the second row gives $-\sum_i p(u_i) \ge (128k - 471)/100 \ge 41/100$. Both contradict the radial bound. Hence every global maximizer has exactly three points in I and nine in O. The row and population checks are replayed by `twelve-point-ward-radii/verify_all.py`. The subsequent pair comparison can therefore use the closures of the three radial products $I \times I$, $I \times O$, and $O \times O$, intersected with the Gram and separation conditions.

4 The twelve-point candidate

We use the mass-11 normalization of Section 2. The candidate is specified by an isolating box for the full force equations. The original box is in mass 12, where the multiplier is $11/2$. In real coordinates the force at the ith point is

$$\mathcal{F}_i(y) = \sum_{j \ne i} \frac{y_i - y_j}{|y_i - y_j|^2} - \frac{11}{2} y_i.$$

Fix the rotational gauge $\operatorname{Im} y_0 = 0$, $\operatorname{Re} y_0 > 0$, and omit the imaginary component of the force at y_0. Let f denote the resulting map on 23 real variables. The certificate specifies a dyadic center m, a closed sup-norm box X of radius $r = 2^{-80}$, and a dyadic preconditioner C. Outward integer interval evaluation verifies that the box is collision free, that the real part of its first point is positive, and that

$$\|Cf(m)\|_\infty \le 2^{-257}, \qquad \sup_{x \in X} \|I - CDf(x)\|_\infty \le 2^{-66}.$$

No.	Layers	t	Mult.	r	No.	Layers	t	Mult.	r
1	3, 3	0.25324601	3	2^{-8}	8	2, 2	−0.53891641	3	2^{-7}
2	3, 3	−0.60361246	6	2^{-8}	9	1, 3	−0.07134427	6	2^{-9}
3	3, 3	0.89560194	3	2^{-10}	10	1, 3	0.42819353	6	2^{-8}
4	2, 3	−1.06454117	6	2^{-9}	11	1, 2	0.21676994	6	2^{-8}
5	2, 3	0.17737052	6	2^{-8}	12	1, 3	−0.35684926	6	2^{-8}
6	2, 3	0.88717064	6	2^{-10}	13	1, 2	−0.43353987	3	2^{-11}
7	3, 3	−1.14884795	3	2^{-11}	14	1, 1	−0.08719201	3	2^{-7}

Table 1: The fourteen unordered pair orbits. Layers are numbered outward, t is rounded for display, and r is the nominal local-box radius used in Section 6. Orbit 13 is on the Gram boundary.

The map $x \mapsto x - Cf(x)$ is a contraction from X into itself. The derivative bound makes $CDf(x)$, hence C, invertible, so its unique fixed point is the unique zero of f in the box. The a posteriori bound $\|Cf(m)\|_\infty/(1-2^{-66})$ encloses this zero within 2^{-256} of m. These data and checks are in `validated-maxima/certificates.json` and `validated-maxima/verify_all.py` at repository revision `3c922f3`.

The identity $\sum_i \operatorname{Im}(\overline{y}_i \mathcal{F}_i) = 0$ follows by grouping the two forces of each pair. It recovers the omitted equation because y_0 is positive real. Summing the full equations gives centering. Taking their real scalar product with the coordinates gives $(11/2)\sum_i |y_i|^2 = 66$. Clearing denominators in the collision-free rational isolating box specifies a unique real solution of a rational polynomial system there, so its coordinates are real algebraic.

The symmetry check uses the smaller a posteriori enclosure. A saved permutation of order three is followed by regauging its first point to the positive real axis. A second saved permutation is treated in the same way after conjugation. The two transformed roots satisfy the force equations and lie in the original uniqueness box. The interval bounds put both transformed roots within 2^{-254} of its center. Uniqueness identifies them with the original root. The permutations generate a dihedral group of order six. The resulting point orbits have sizes $3, 3, 6$ and their squared-radius intervals are disjoint. The symmetry permutations and their replay are recorded in `candidate-global-screen/`.

Rescale by $\sqrt{11/12}$, and rotate a point of the innermost triangle to the positive real axis. Write $\omega = e^{2\pi i/3}$. The exact candidate z^* is the union of the four orbits

$$\{a\omega^j\}_{j=0}^2, \qquad \{-b\omega^j\}_{j=0}^2, \qquad \{(c+id_0)\omega^j\}_{j=0}^2, \qquad \{(c-id_0)\omega^j\}_{j=0}^2, \tag{17}$$

with $a, b, c, d_0 > 0$. The certified reflection fixing the inner point gives the real axis in this gauge. Its fixed point on the middle triangle is negative real, and the outer orbit has a representative with both coordinates positive. These signs are checked on the root enclosure. In particular, $a^2 \in (17/100, 18/100)$ and $b^2 \in (107/100, 109/100)$. The orbits of unordered pairs give fourteen distinct triples (u, v, t) after ordering the radii. Thirteen satisfy $t^2 < uv$. The remaining triple comes from the opposite inner and middle rays and has $t = -\sqrt{uv}$. The orbit multiplicities sum to 66. The squared radii in mass eleven are approximately 0.17438402, 1.07783282, and 1.20722491. Table 1 lists the contact orbits in certificate order.

For later use, the certificate also encloses the third moment of each radius layer in the axis gauge. The two triangle moments have the moduli dictated by their radii. On the outer layer, with squared radius $s_3 = c^2 + d_0^2$, it verifies

$$0 < \left| \sum_{|z_i^*|^2 = s_3} (z_i^*)^3 \right| < 6 s_3^{3/2}.$$

The interval computation can use $(z_i^* \overline{w})^3$ for an inner axis point w, and compare squared moduli before dividing out the common factor $|w|^6$. This test separates the outer third moment from zero and from the endpoint where the two cubed representatives coincide. The layer tests are in `candidate-global-screen/symmetry_report.json`.

If $w_1, \dots, w_4$ are the cubes of the four representatives in (17), the candidate value in mass 12 is

$$C_{12} = \left(\frac{12}{11}\right)^{66} 27^4 \prod_{i=1}^{4} |w_i|^2 \prod_{i<j} |w_i - w_j|^6. \tag{18}$$

The factor $z^3 - w_i$ contributes $27|w_i|^2$ within its triangle. The squared modulus of the resultant of $z^3 - w_i$ and $z^3 - w_j$ gives $|w_i - w_j|^6$ between triangles.

5 Exact contact interpolation

Set $B = \log \Delta(z^*)$ in mass 11 and use the diagonal correction of Section 2. Put $u_i^* = |z_i^*|^2$ and denote the three distinct squared radii by s_1, s_2, s_3. Define

$$\mu_j = \frac{1}{12} \sum_i (u_i^*)^j, \qquad q(u) = \prod_{r=1}^{3} (u - s_r), \qquad F_0(u) = (u - \mu_1, u^2 - \mu_2, u^3 - \mu_3, u^4 - \mu_4, 10q(u)^2)^T.$$

The radial feature sum at the candidate is zero. For the other blocks use $\phi_j(u) = (u-1)^j$, $0 \le j \le 4$, and write $M_k^* = \sum_i (z_i^*)^k \phi(u_i^*)$ for the raw moment vector. For $1 \le k \le 11$ with $3 \nmid k$, use the five raw features. Their candidate moments vanish by the rotation of order three. For $k = 3, 6, 9$, use the four features

$$\phi_j(u) - \frac{(M_k^*)_j}{(M_k^*)_0}, \qquad 1 \le j \le 4.$$

The ratios are real by reflection in the chosen axis. The root interval verifies all three denominators are nonzero. Each adapted feature sum is therefore zero in its harmonic. There are nine symmetric Gram blocks of size five and three of size four, giving 165 independent Gram entries. The ordered-pair sum of the base kernel and its first differential vanish at z^* for every choice of these entries.

5.1 First-order restrictions on the Ward correction

Use the Ward blocks of Section 2, with the vector parameter in $A_k = v_k a_k^T + a_k v_k^T$ denoted by a_k. For $k = 0, 3$ impose

$$a_0^T M_0^* = a_3^T M_3^* = 0. \tag{19}$$

Together with $v_k^T M_k^* = 0$, these conditions give $A_k M_k^* = 0$. The corresponding products for $k = 1, 2$ vanish because $M_1^* = M_2^* = 0$. Thus the Ward ordered-pair sum also has zero first differential at the candidate. Each condition in (19) eliminates the constant coordinate of its vector parameter. With nine parameters in A_1, five in A_2, and four in each of A_0, A_3, the Ward correction has 22 parameters.

These two eliminations can also be read directly from sharp contact. All candidate pairs are strict in the radius and distance bounds. Small variations of the actual points preserve the Gram condition, including at the collinear pair. If $G = h - \log d$ is nonnegative there and vanishes at

every candidate pair, the sum of its values has zero first differential. Along the dihedral family (17), the differential of that sum is

$$-(a_0^T M_0^*)\,\mathrm{d}Q - (a_3^T M_3^*)\,\mathrm{d}W_3, \qquad W_3 = \sum_i (|z_i|^2 - 5/3) z_i^3.$$

Indeed $v_0^T M_0 = Q - 11$, $v_3^T M_3 = W_3$ is real in this family, and the force equations give $\mathrm{d}\log\Delta = 6\,\mathrm{d}Q$ at the candidate. The variation $\dot a = b$, $\dot b = -a$, $\dot c = \dot d_0 = 0$ has $\mathrm{d}Q = 0$ and

$$\mathrm{d}W_3 = 15ab\big[a(a^2-1) + b(b^2-1)\big] < 0.$$

The sign follows from the two certified radius intervals. This variation forces $a_3^T M_3^* = 0$. A variation with nonzero $\mathrm{d}Q$ then forces $a_0^T M_0^* = 0$.

5.2 The five compatibility relations

Let $\theta \in \mathbb{R}^{187}$ contain the Gram and Ward entries. Write

$$h = h_0 + \mathcal{H}_\theta, \qquad h_0 = \frac{6}{11}(u+v) + \frac{B-66}{66},$$

so that $\mathcal{H}_\theta$ is linear in θ. At each contact impose the value condition $G = 0$. At the thirteen interior contacts impose $\nabla G = 0$. At the boundary contact impose the two derivatives of $G(u, v, -\sqrt{uv})$ to be zero. These are $G_u + tG_t/(2u) = 0$ and $G_v + tG_t/(2v) = 0$. Six interior contacts have $u = v$, where exchange symmetry identifies G_u and G_v. After dropping these duplicates there are 14 value rows and 35 derivative rows. Write the resulting system as $L\theta = y$, where y contains the corresponding values and derivatives of $\log d - h_0$.

For explicit matrix construction, let K'_θ be the sum of the base kernel and its Ward correction. The column for a parameter is obtained by evaluating and differentiating its contribution to

$$\mathcal{H}_\theta(u,v,t) = -K'_\theta(u,v,t) - \frac{K'_\theta(u,u,u) + K'_\theta(v,v,v)}{22}.$$

In particular the derivatives of the two diagonal terms enter the radial rows. At each contact the derivative of $\log d$ is $(1, 1, -2)/d$, so only the value rows require logarithms on the right-hand side. The entries of L are algebraic.

One row relation is the sum of the value rows with the pair-orbit multiplicities as weights. It vanishes on $\mathcal{H}_\theta$ because the candidate kernel sum is zero. It also vanishes on the right-hand side, since the weighted sum of $\log d$ is B and that of h_0 is B.

Four more relations come from varying a, b, c, d_0 in (17). In each direction apply the chain rule to the multiplicity-weighted sum over the contacts. The derivative of the kernel sum is zero by the adapted features and (19). On the right-hand side, both $\log\Delta$ and the sum of h_0 have differential $6\,\mathrm{d}Q$. On the boundary orbit, the variation stays on $t = -\sqrt{uv}$, so the chain rule uses the two retained tangent rows. On an equal-radius orbit it uses the retained radial row with the sum of the two radial velocities.

The four derivative relations are independent. A variation giving the zero row combination has zero contact velocities in all the retained coordinates, since each orbit multiplicity is positive. In particular it has zero differentials of a^2, b^2, $c^2 + d_0^2$, and the inner–outer dot product ac. These four differentials are independent because a, b, d_0 are positive. The value relation is supported on different rows. Hence the five relations are independent and $\operatorname{rank} L \le 44$.

5.3 Certified rank and correction of the seed

The stored row and column sets I, J specify a 44×44 minor. Rebuilding its entries from the isolated root and applying outward interval Gaussian elimination on the 2^{-384} lattice verifies that every pivot excludes zero and that the determinant interval excludes zero. Thus $\operatorname{rank} L = 44$, and the five analytic identities give all row compatibility conditions. The row and column sets are recorded in `twelve-point-sharp-contacts/minor.json`.

The global certificate takes the dyadic Gram and Ward seed from `twelve-point-banded-upper/kernel.json`. Each harmonic Gram block is transferred through its stored change-of-basis matrix. For $k = 3, 6, 9$ the transfer uses the last four rows, since the constant coefficient of each adapted feature is supplied by the exact candidate moment ratio. The radial features use the exact nodes and means, and the constant coordinates of a_0, a_3 are supplied by (19). Denote the resulting seed in the 187 coordinates by $\theta^{(0)}$. The exact corrected coefficients are defined by

$$\theta_J = \theta_J^{(0)} + L_{IJ}^{-1}(y_I - (L\theta^{(0)})_I), \qquad \theta_{J^c} = \theta_{J^c}^{(0)}. \tag{20}$$

The selected equations hold by construction. The row relations extend them to all 49 equations, because y satisfies those relations. This specifies the coefficients as exact real expressions in the algebraic candidate data and the contact logarithms.

For interval evaluation, each contact distance is scaled by a power of two into $[1, 2]$. With $r = (s-1)/(s+1)$ for the scaled distance s, the logarithm enclosure uses $2\sum_{j=0}^{99} r^{2j+1}/(2j+1)$ and the remainder bound $2r^{201}/(201(1-r^2))$. The same series encloses $\log 2$. The replay encloses the solve in (20), checks all contact residual intervals, and applies interval LDL^T factorization to the twelve corrected Gram blocks. Every Gram block is strictly positive definite, with all pivots greater than 2^{-17}. It also certifies positive Hessians at the thirteen interior contacts and for the restriction to the Gram face at the boundary contact, with $-tG_t > 2^{-13}$ there. The reconstruction is implemented in `twelve-point-sharp-global/kernel_bridge.py` and recorded in its verification report.

6 Zero-slack coverage of the pair domain

Work at mass 11, and let z^* denote the isolated candidate. Use the function h from Section 2, with the coefficients reconstructed in Section 5 and with $B = \log \Delta(z^*)$. Write

$$d = u + v - 2t, \qquad G(u, v, t) = h(u, v, t) - \log d.$$

By Section 3, every pair in a global maximizer has its squared radii in the two bands $I = (3/25, 6/25)$ and $O = (47/50, 71/50)$. We use their closures in the cover. Since h and d are symmetric in u, v, we order the radii by $u \le v$. The remaining constraints are $t^2 \le uv$ and $d \ge 1/3$. The three radial rectangles are $\overline{I}^2$, $\overline{I} \times \overline{O}$, and $\overline{O}^2$. For these rectangles take, respectively, $R = 6/25, 3/5, 71/50$, so that $\sqrt{uv} \le R$. Appending the intervals $[-R, 0]$ and $[0, R]$ gives six rational root boxes.

6.1 The fourteen contact neighborhoods

For an exact contact $p_* = (u_*, v_*, t_*)$, put $d_* = u_* + v_* - 2t_*$ and $s = (d - d_*)/d_*$. The polynomial

$$P_*(p) = h(p) - \log d_* - \left(s - \frac{s^2}{2} + \frac{s^3}{3}\right) \tag{21}$$

is a lower bound for G wherever $d > 0$. Indeed, the derivative of $s - s^2/2 + s^3/3 - \log(1+s)$ is $s^3/(1+s)$. The difference therefore has its minimum zero at $s = 0$ on $s > -1$. In coordinates translated by p_*, the constant coefficient of P_* is exactly zero by the contact equation. The linear term agrees with that of G. The checker inserts the exact zero constant and encloses the other translated coefficients by outward rounding on the 2^{-384} dyadic grid.

For each of the thirteen interior contacts, the local data supply a dyadic radius r and a dyadic 3×3 congruence matrix W. On the full cube of offsets $[-r, r]^3$, the checker bounds the polynomial entries of $W^T \nabla^2 P_* W$ and verifies positivity by interval LDL^T elimination. The resulting positive definite congruence implies that W is invertible and that $\nabla^2 P_*$ is positive definite throughout the cube. The cube lies in $d > 0$. Since $P_*(p_*) = 0$ and $\nabla P_*(p_*) = 0$, integration of the second derivative along the segment from p_* to p yields $P_*(p) \geq 0$, with equality only at p_*.

The remaining contact is antiparallel and lies on the Gram face $t = f(u, v) = -\sqrt{uv}$. Here the proof box has half-widths $(r, r, 2r)$. The checker verifies $u, v, d > 0$ on this box, verifies that the projection $\psi(u, v) = (u, v, f(u, v))$ lies in it for every base point, and proves $\partial_t P_* > 0$ throughout the box. On the face it bounds the two-dimensional Hessian

$$\nabla^2 (P_* \circ \psi) = D\psi^T (\nabla^2 P_*) D\psi + (\partial_t P_*) \nabla^2 f.$$

The last row of $D\psi$ is $(f/(2u), f/(2v))$. The diagonal entries of $\nabla^2 f$ are $-f/(4u^2)$ and $-f/(4v^2)$, and the off-diagonal entry is $f/(4uv)$. Interval LDL^T verifies positive definiteness over the full base rectangle. The exact tangential contact equations make p_* the unique minimum on this projected rectangle. Every feasible point in the proof box satisfies $t \geq f(u, v)$, and the vertical segment from the face to that point lies in the box. Positivity of $\partial_t P_*$ thus extends $P_* \geq 0$ from the face to the feasible side. Equality again occurs only at the contact.

The replay verifies all fourteen local boxes. Their smallest nominal radius is 2^{-11}, and the lower bound 2^{-9} applies to the pivots in the verified local congruences and the boundary restriction. To assign a rational cover leaf to a contact, the checker uses a rational box contained in the certified box about the exact center. Specifically, if a center coordinate is enclosed by $[L, U]$, it uses $[U - r, L + r]$ in that coordinate, and checks that this interval has positive length. This is the intersection of the radius-r intervals over all centers in $[L, U]$. The same nominal r is used for the boundary contact's rational assignment box.

6.2 The signed Bernstein cover

Write $h = \sum c_{ijk} u^i v^j t^k$, with the real coefficients enclosed by rational intervals. On each signed root box we construct a rational polynomial lower envelope for h. For a nonnegative monomial we use the lower coefficient endpoint. For an odd power of t on the nonpositive half, we use the upper endpoint. The signs are fixed because $u, v \geq 0$. This produces a pointwise lower envelope on the whole root box.

The envelope is converted to the tensor Bernstein basis of degrees $(19, 19, 11)$. To describe the conversion, first map the box affinely to the unit cube. If $\widetilde{c}_\alpha$ are the resulting power coefficients and $m = (19, 19, 11)$, the control coefficient at multi-index r is

$$\beta_r = \sum_{\alpha \leq r} \widetilde{c}_\alpha \prod_{\ell=1}^{3} \frac{\binom{r_\ell}{\alpha_\ell}}{\binom{m_\ell}{\alpha_\ell}}.$$

The initial conversion uses exact rationals. Coefficients are rounded downward on the 2^{-62} grid, and midpoint de Casteljau subdivision uses downward rounding at each average. The resulting

controls remain lower bounds after every restriction. The implementation stores the controls in 96-bit lanes of a Python integer with a 2^{90} bias. The checked lane ranges keep the relevant sums and comparisons separate, so that the packed operations implement the same componentwise inequalities.

For a box with distance endpoints $d_{\mathrm{lo}}, d_{\mathrm{hi}}$, the usual comparison uses the tangent

$$\log d \le \log a + \frac{d-a}{a}, \qquad a = \frac{\max(1/3, d_{\mathrm{lo}}) + d_{\mathrm{hi}}}{2}.$$

Only the value at the rational tangent center requires a logarithm enclosure. The checker reduces $a = 2^e y$ with $1 \le y < 2$ and uses the first 24 terms of the series

$$\log y = 2 \sum_{j \ge 0} \frac{q^{2j+1}}{2j+1}, \qquad q = \frac{y-1}{y+1}.$$

For $0 \le q \le 1/3$, the omitted tail is bounded above by $3q^{49}/49$. All series operations are rounded outward on the 2^{-96} grid. The sign of e determines which endpoint of the enclosure for $\log 2$ enters the upper bound. The affine comparison is then rounded upward in Bernstein coordinates.

Some leaves use a quadratic comparison with $a = d_{\mathrm{hi}}$. For $x = (a-d)/a$ and $0 \le x < 1$, $\log d \le \log a - x - x^2/2$. The deficit x is a sum of three nonnegative affine coordinate deficits. Consequently the linear, square and cross-product contributions to the Bernstein coefficients of $x + x^2/2$ can be rounded downward before subtraction from the upper constant. For the cross products the implementation multiplies two nonnegative factors rounded down on the 2^{-31} grid. Either comparison is accepted when every lower control of h dominates the corresponding upper control of the logarithm majorant. The Bernstein basis is nonnegative and sums to one, which yields $G \ge 0$ on the admissible part of the leaf.

Each action is a midpoint split, an accepted logarithm comparison, containment of the entire box in a certified contact box, or exclusion of the box. Exclusion uses one of three rational tests. The box lies in $u > v$, its minimum $|t|^2$ exceeds $u_{\mathrm{hi}} v_{\mathrm{hi}}$, or its distance upper bound is below $1/3$. The fixed action streams are checked as complete prefix trees, and each action is replayed with the corresponding box and control tensor. Table 2 gives the resulting counts. Here A counts logarithm comparisons, L counts contact-box leaves, and P counts exclusions.

Root domain	Nodes	A	L	P
$I, I,\ t \le 0$	7 595	3 403	112	283
$I, I,\ t \ge 0$	39	17	0	3
$I, O,\ t \le 0$	144 183	53 620	1 276	17 196
$I, O,\ t \ge 0$	93 415	45 990	408	310
$O, O,\ t \le 0$	47 105	20 721	312	2 520
$O, O,\ t \ge 0$	435 713	211 754	513	5 590
Total	728 050	335 505	2 621	25 902

Table 2: Exact zero-slack replay of the six signed root boxes.

The last row of the six root domains uses 3 709 quadratic comparisons within its count A. The maximum depth is 52. There are 364 022 splits, 364 028 terminal boxes and zero pending boxes. The node and leaf conservation identities are checked separately for each root. The data and checker are in `twelve-point-sharp-global` at revision `3c922f3`. The complete replay uses the Python standard library and has pair slack exactly zero.

Every pair of a global maximizer z now satisfies $h \geq \log d$. Summation and the stationary kernel identity of Section 2 give

$$\log \Delta(z) \leq \sum_{i<j} h_{ij} = B - \tfrac{1}{2} \sum_{i,j} K_{ij} \leq B.$$

The candidate attains B in the mass-11 normalization. Scaling back to mass 12 proves $M_{12} = C_{12}$.

7 Equality and the maximizing configuration

At equality the sum of the base kernel K vanishes. Let F_k denote the radial feature vector in harmonic k and set $T_k = \sum_i z_i^k F_k(u_i)$. The ordered kernel sum is $\sum_{k=0}^{11} T_k^* Q_k T_k$. The twelve Gram blocks are positive definite by Section 5, so $T_k = 0$ for every k.

Let $\alpha_1, \alpha_2, \alpha_3$ be the candidate's three distinct squared radii and $q(u) = \prod_r (u - \alpha_r)$. The radial feature $10q(u)^2$ gives $\sum_i q(u_i)^2 = 0$, and therefore every u_i is one of the three nodes. The population N_r at node α_r satisfies the same zeroth, first and second radial moment equations as the candidate. The matrix $(\alpha_r^j)_{0 \leq j \leq 2, 1 \leq r \leq 3}$ is Vandermonde and invertible. Thus $(N_1, N_2, N_3) = (3, 3, 6)$.

For $k = 1, 2, 4, 5$ the features span the radial polynomials of degree at most four. In particular they span the degree-two Lagrange polynomial that equals one at α_r and zero at the other two nodes. The condition $T_k = 0$ gives the corresponding power sum zero on each layer separately. Newton's identities give $z^3 + A_r$ as the root polynomial of each three-point layer. On the six-point layer the first, second, fourth and fifth elementary symmetric functions vanish, giving the root polynomial $z^6 + Az^3 + D$. Its two roots as a polynomial in z^3 are distinct because the maximizing configuration is collision-free. Hence the outer layer consists of two equilateral triangles on the same circle.

To recover the relative phases, write

$$S_r = \sum_{i : u_i = \alpha_r} z_i^3, \qquad M_j = \sum_r (\alpha_r - 1)^j S_r, \quad 0 \leq j \leq 4,$$

and use stars for the candidate moments. The four third-harmonic features give $M_j = (M_j^\star / M_0^\star) M_0$ for $1 \leq j \leq 4$. The denominator $M_0^\star$ is certified nonzero. The equations for $j = 0, 1, 2$ and a second Vandermonde inversion give

$$S_r = \zeta S_r^\star \quad (1 \leq r \leq 3), \qquad \zeta = M_0 / M_0^\star.$$

On a three-point layer the cubic sum has modulus $3\alpha_r^{3/2}$, so $|\zeta| = 1$. A common rotation through θ multiplies all cubic sums by $e^{3i\theta}$, so a rotation can set $\zeta = 1$. The two three-point layers are then fixed, since the common cubic value on each is $S_r/3$.

On the outer layer the two cubic values w_+, w_- have common modulus $R_3 = \alpha_3^{3/2}$ and sum $W_3 = S_3/3$. The certified candidate has $W_3 \neq 0$. Writing $w_\pm = W_3/2 \pm q_3$, equality of their moduli gives $\mathrm{Re}(\overline{W_3} q_3) = 0$, and their common modulus fixes $|q_3|^2 = R_3^2 - |W_3|^2/4$. These two conditions determine the unordered pair. Taking all cubic roots of the two values recovers the outer layer. The maximizing configuration therefore coincides with z^* after rotation and relabeling. Together with the exact value formula (18), this completes the proof of Theorem 1. □

8 Certificate data and reproduction

The finite inputs are fixed at Git commit `3c922f3446f4f32441d75aef1d7907d38b0544f7`. Each certificate's manifest fixes the exact input bytes by SHA-256. The replay programs reconstruct the

quantities in the proofs and check their signs or inclusion conditions. The archived reports give the counts and lower bounds obtained by those programs.

Archive directory	Proof data and verification task
`validated-maxima` and `candidate-global-screen`	Dyadic stationary root certificate, symmetry permutations, contact multiplicities and third-moment nondegeneracy.
`twelve-point-removal-domain`	Integer witness and exact inequalities for the deleted-point bounds.
`twelve-point-ward-upper`	Rational Gram blocks and Ward parameters, with a 268,239-node cover for the preliminary pair budget.
`twelve-point-ward-radii`	The 53 sets of row multipliers and their 99,603-node cover, and the radial dual vector with its exact matrix equation.
`twelve-point-sharp-contacts` and `twelve-point-banded-upper`	The selected contact minor and the dyadic seed used in the final coefficient reconstruction.
`twelve-point-sharp-global`	Exact reconstruction, fourteen local box tests, and six zero-slack trees with 728,050 nodes in total.

From the root of the pinned archive, the following commands replay the analytic lemmas' finite hypotheses in dependency order.

```
python -B -S -O twelve-point-removal-domain/verify_all.py
python -B -S -O twelve-point-ward-upper/verify_all.py
python -B -S -O twelve-point-ward-radii/verify_all.py
python -B -S -O twelve-point-sharp-global/verify_all.py
```

The last command also reconstructs the certified candidate, symmetry, and contact minor. It accepts the preceding radial theorem as an input. The archive's `scripts/verify_current.py` provides a broader dependency-ordered replay. The core proof programs use the Python standard library. The sharp verifier supports `-jobs 1` for sequential replay. The semidefinite solver and the Jacobian exact linear-system tool contributed to discovery. For example, the saved Jacobian solution for the radial dual vector is checked by direct rational substitution.

9 Construction principles for other point counts

9.1 Use an upper-bound gap to constrain individual points

The pair inequality and the ordered Gram sum give two separate nonnegative terms in (9). Keeping both is what yields geometry from a preliminary objective bound. For a marked point with squared radius u, centering and mass $n-1$ give

$$\sum_{j\neq i}\left(v_j-\frac{n-1-u}{n-1}\right)=0,\qquad \sum_{j\neq i}\left(t_{ij}+\frac{u}{n-1}\right)=0.$$

On a proposed excluded interval, seek a pointwise lower bound for the pair deficit by a positive constant plus these two affine directions. After summing the row, the affine terms cancel. Comparing the constant with the global budget certifies exclusion of the entire interval. Once the surviving bands are separated, test integer band populations using a dual vector for the radial Gram block and simple one-variable supporting inequalities. This step uses the raw population constraints and the moment penalty in different ways. It converts the budget into a small collection of necessary pair boxes without assigning a symmetry group to the optimizer.

9.2 Build the kernel around the candidate's vanishing moments

For a candidate with cyclic symmetry of order m, the angular feature moments vanish for every harmonic k with $m \nmid k$, regardless of the radial weights. At multiples of m, replace the raw feature space by the orthogonal complement of the candidate's moment direction. The resulting free Gram blocks can be strictly positive definite in the reduced feature spaces. For radial features, subtract the candidate's power means and include the square of the polynomial that vanishes at its radius nodes. At equality the squared polynomial fixes each individual radius, while the power means fix the populations.

Design the remaining angular features with equality in mind. For layers of m or $2m$ points, vanishing layer power sums for $1 \le k < 2m$, $k \ne m$, make each layer polynomial a polynomial in z^m. Its roots are one or two regular m-gons. A common mth-moment direction fixes the relative phases when an m-point layer anchors the overall rotation and each double layer has nonzero sum of its two mth powers. For candidates with a common reflection, the moment directions are real in an axis gauge and the kernel can use the three real pair variables u, v, t.

9.3 Use the force identities as extra kernel parameters

The low-order Ward directions give freedom in the pointwise majorant while leaving its ordered sum unchanged on stationary configurations. There is also a higher-order construction. In mass $n-1$ let

$$v_k = \left(-\frac{n-k-1}{n}, 1, 0, 0, 0\right)^T \quad (k \ge 4).$$

Equation (4) and $p_1 = 0$ give

$$v_k^T M_k = \frac{1}{n} \sum_{r=2}^{k-2} p_r p_{k-r}.$$

For a real vector a, take $A_k = v_k a^T + a v_k^T$. If $u \le U$ on the necessary domain, take L_k with

$$L_k \ \ge \ \max_{0 \le u \le U} u^{k/2-1} |a^T \phi(u)|.$$

Then $|a^T M_k| \le (n-1) L_k$. Applying $2|p_r p_{k-r}| \le |p_r|^2 + |p_{k-r}|^2$ in the rank-two form gives

$$M_k^* A_k M_k \ge -\frac{2(n-1)L_k}{n} \sum_{r=2}^{k-2} |p_r|^2.$$

Since the ordered sum of H_r equals $|p_r|^2$, the pair function

$$H_k(u,v,t)\phi(u)^T A_k \phi(v) + \frac{2(n-1)L_k}{n} \sum_{r=2}^{k-2} H_r(u,v,t)$$

has nonnegative ordered sum on the stationary domain. For a candidate with m-fold rotational symmetry it has zero ordered sum when $4 \le k \le m+1$. For an even k one can bound L_k by linear inequalities on univariate Bernstein coefficients. This gives additional variables for a future search with the same pair summation identity after its diagonal correction is recomputed.

9.4 Separate interpolation, local sign, and global coverage

Use a semidefinite search to find a promising kernel under sampled pair inequalities and affine contact conditions. Derive the exact linear dependencies of the contact system by summing pair values and differentiating along candidate variations. Then fix a nonzero minor, retain the exact logarithms of the contact distances in the right-hand side, and solve for the corresponding correction of the seed. Positive pivots certify that the adjusted Gram blocks retain their signs.

Exact contact generally calls for coefficients containing logarithms. For the algebraic candidate in Section 4, a polynomial with algebraic nonconstant coefficients would have an algebraic difference between its values at any two contacts, even if its constant term were allowed to be arbitrary. For two different squared distances the required difference is $\log(d_1/d_2)$, which is transcendental by the Hermite–Lindemann theorem [3]. The reconstruction therefore uses the individual contact logarithms in the coefficient vector.

For the pair inequality, use a contact-centered polynomial upper bound for the logarithm and a Hessian test where the gap must vanish. Treat Gram-boundary contacts in tangential and inward directions separately. Cover the complementary region by subdivision, returning failed boxes to the numerical search as new constraints. A proof at another point count requires its own exact contact system, positive Gram blocks, local sign tests, and complete cover of the necessary domain.

References

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